

Giorgio Caizzi

giocaizzi@gmail.com | Milan, Italy

 giorgio-caizzi  giocaizzi

Engineer who builds and owns production AI systems end to end — the retrieval, agents, event-driven backends, CI/CD, observability and IaC around them. Self-taught into code at AECOM, promoted from consultant to tech lead in two years, now shipping GenAI for banks, media and retail.

EXPERIENCE

Associate Manager, Products (prev. Senior Consultant)

May 2024 - Present

Jakala S.p.A.

Milan, Italy

- Built a production RAG over 600,000 pages of internal banking regulation (PDF, Word, Excel) as lead engineer on a year-long embedded Banca Generali engagement — structure-aware chunking, hierarchical retrieval, version-aware metadata and an event-driven sync pipeline into the bank's platforms, quality-gated by a golden evaluation set. In production, rolling out across 1,600 employees.
- Built a governed tool layer that lets AI agents run SQL and geospatial operations — a GraphQL wrapper with RBAC constraining each agent call to the user's role. Non-technical users chat with their data beside a live map; productized for clients including Regione Lombardia and NielsenIQ.
- Led the engineering team that built an AI pipeline generating on-model product photography for Benetton (ComfyUI, Flux, LoRAs on Azure GPUs) at up to 90 SKUs/day with hybrid human-and-AI QA — replacing studio photoshoots for most of the brand's ecommerce imagery by autumn 2025.
- Developed the vision-language pipeline that estimates a meal's carbohydrates from a single photo for T.Conto (a consumer nutrition app live on the App Store and Google Play), using agents to retrieve ingredient reference values from a vector store of the official food-composition tables.

Environmental Engineer

Feb. 2021 - May 2024

AECOM URS Italia S.p.A.

Milan, Italy

Remediation of contaminated industrial sites for ENI, Q8 and the US DoD.

- Built Python pipelines (geopandas, GDAL, rasterio, PostGIS) that turned days of manual GIS work into hours — open-sourced on GitHub as [py3dinterpolations](#).

Brand Activation & Live Events

Nov. 2013 - Feb. 2021

Red Bull, GoPro EMEA

Italy & Europe

Seven years of live brand-activation at major sporting events across Europe, alongside my studies — public speaking and running games/giveaways for crowds, trilingually.

EDUCATION

MSc (102/110) and BSc (97/110) in Environmental & Land Planning Engineering

Milan, Italy

Politecnico di Milano

Sep. 2014 - Dec. 2020

Relevant coursework: computer science (30/30), GIS & spatial data, statistical models & stochastic processes, remote sensing

Theses: PM2.5 spatial-interpolation algorithm (Wiseair S.r.l.); facility-location optimization for Milan's public water network (MM S.p.A.)

The British School of Brussels

Brussels, Belgium

International Baccalaureate Diploma

Sep. 2012 - May 2014

TECHNICAL SKILLS

GenAI & LLMops: RAG & retrieval engineering, LLMs & multimodal, agentic orchestration & MCP, evaluation & guardrailing, AI observability, OpenAI, Anthropic & Azure OpenAI

Backend & Systems: Python & TypeScript, FastAPI, REST & GraphQL, event-driven pipelines & queues, OpenTelemetry distributed tracing, Docker, CI/CD

Data: SQL & NoSQL, Redis, vector & graph stores

Cloud & Infrastructure: Azure, GCP, Terraform (IaC), networking, serverless

Geospatial: PostGIS, GDAL, geopandas, shapely, rasterio, ArcGIS & QGIS

PROJECTS

 **ralph-copilot** autonomous agent loop for GitHub Copilot ★138

 **greenhouse** smart-irrigation app exposing an MCP server to LLM agents

 **btc-cycles** bitcoin-cycle analytics with a live daily dashboard

Copilot custom agents, git-as-memory

Python, FastAPI, MCP, Docker

Python, PyPI, CI/CD, plotly

PERSONAL SKILLS

Languages: Italian (native), English (bilingual), French (professional working proficiency).

Interests: Surfing, skiing and road-cycling, chased worldwide: Indonesia, Angola, South Africa, across Europe.